

System of lin eqⁿ

shorthand: $Ax=b$

"easy" opⁿ on equations

"Equivalent" system of eqⁿ
ie. solⁿ set does not change

dictionary

eqn $\leftrightarrow$ rows
var $\leftrightarrow$ col^s of A

Augmented matrix $[A|b]$

row opⁿ
(reversible)

$\begin{cases} R_i + aR_j \\ R_i \leftrightarrow R_j \\ cR_i \quad c \neq 0 \end{cases}$

Convert to REF

Defⁿ $r \times c$ Matrix A is in REF if the following is true for all $2 \leq i \leq r$.

If (row i has a nonzero entry) then (so must Row $i-1$ and left most nonzero entry in row $i-1$ must be in an earlier column than "that" in row i).

We proved: Every matrix can be turned into a matrix in REF by doing row opⁿ (pf by induction on number of rows)

$$\left[\begin{array}{cccc|c} 2 & 4 & 6 & 8 & 1 \\ 3 & 0 & 7 & 1 & 10 \end{array} \right] \longrightarrow \left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & \frac{1}{2} \\ 0 & 1 & \frac{2}{6} & \frac{11}{6} & \frac{17}{12} \end{array} \right] \longrightarrow \left[\begin{array}{cccc|c} 1 & 0 & \frac{7}{3} & \frac{1}{3} & \frac{20}{6} \\ 0 & 1 & \frac{1}{3} & \frac{11}{6} & -\frac{17}{12} \end{array} \right]$$

REF

Claim:

$Ax=b$ is inconsistent $\iff$ REF of $[A|b]$ has a pivot in last column

Pf. $\Leftarrow$ is easy. For $\Rightarrow$ we prove contrapositive i.e. $\left\{ \begin{array}{l} \text{if no pivot in last col then} \\ Ax=b \text{ is consistent} \end{array} \right.$

Suppose there is no pivot in last col. Then
each pivot variable = unique expression in terms of
subsequent variables.

Note For a matrix in REF
each row can have at most one pivot.
each col ———— //

General shape of REF

0	...	0		1		*	...	*	...	*				
						0	...	0		1		*	...	*
										0				

pivot column = column in which
a pivot (in REF)
occurs

Given REF of augmented matrix of $Ax=b$
without pivot in the last column,

I obtain a unique solution to $Ax=b$
by assigning arbitrary values to variables
corresponding to non-pivot columns.
(we call these "free variables")

Consequences If $Ax=b$ is consistent

it has a unique solⁿ

$\Leftrightarrow$ there is no free variable in $\text{REF}(A|b)$

$\Leftrightarrow$ Every col except last of $\text{REF}(A|b)$
has a pivot

↑
Formally proof by induction on
of nonzero rows, working
from last eqⁿ to first.

General solution of example using RREF =

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = x_3 \begin{bmatrix} -\frac{7}{3} \\ -\frac{1}{3} \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -\frac{1}{3} \\ -\frac{11}{6} \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} +\frac{20}{6} \\ -\frac{17}{6} \\ 0 \\ 0 \end{bmatrix}$$